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Fully distributed consensus of linear multi-agent systems via dynamic event-triggered control ☆

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ABSTRACT

This paper investigates the fully distributed consensus control problem for general linear multi-agent systems via dynamic event-triggered scheme. To address the need to reduce network communication burdens and avoid involving any global features of multi-agent networks, a novel fully distributed controller based on the open-loop state estimator of agents is presented. To improve anti-interference ability and practicability of the proposed approach, an event-triggered strategy which can guarantee a precise lower bound of triggering intervals for each follower is designed with a virtual internal variable. Then, the stability of the multi-agent systems is analyzed without any global network information. Several simulation experiments are conducted to illustrate theoretical results.

1. Introduction

Over the past two decades, abundant attention has been devoted by scholars to the cooperative control of multi-agent systems, which has found practical applications in various domains such as formation control of unmanned aerial vehicles (UAVs) [1], distributed sensor networks [2], spacecraft coordination [3] and cooperative surveillance [4]. Among numerous branches of the cooperative control of multi-agent systems, consensus is a basic and crucial concept. Its objective is to make all agents reach the same state through effective controllers based on information exchange with their neighbors. Initially, consensus problems focused mainly on leaderless multi-agent systems [5–7]. To achieve a preset goal and facilitate flexible control of convergence trajectories, the concept of a leader was introduced into multi-agent systems, increasing the development of tracking control (multi-agent systems with a leader) [8–10] and containment control (multi-agent systems with two or more leaders) [11–13].

Traditional consensus control algorithms typically require continuous communication among agents, leading to increased hardware requirements. However, practical applications are often limited by computing costs, communication delays, bandwidth, and excessive power consumption. These limitations make it challenging to shorten communication time among agents in consensus problems. Up to now, abundant efforts have been made to deal with communication limitations, and several control strategies have been put forward, such as intermittent control [14], sampled-data control [15–20], and event-triggered control. It should be pointed out that the sampled-data control needs to follow fixed sampling time rules to obtain neighborhood information at a specific moment and transmit it to the controller. This rigid sampling law often leads to sampling actions that do not significantly improve convergence efficiency and requires all agents to exchange information synchronously after sampling, resulting in resource consumption that makes sampled-data control unsuitable for actual control systems.

As an appealing alternative, the event-triggered strategy sets flexible sampling conditions to ensure system stability and improve system convergence efficiency [21,22]. More recently, numerous researchers have devoted efforts to event-triggered control. In 2011, a controller with a centralized decision maker based on event-triggered mechanism was designed [21]. Although agents do not need to update controllers continuously, they need to monitor state errors of all agents and update controllers simultaneously. To overcome these limitations, a decentralized event-triggered controller was developed [21]. This idea was later extended to general linear systems [23,24], nonlinear multi-agent systems [22] and even synchronization of coupled dynamical systems [25]. To relieve communication costs, the event-triggered approach has also been combined with sampled-data control [26–28].

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Meng and Chen [26] proposed an event-triggered consensus algorithm based on sampled-data detection. Then, it was extended to the consensus problem of nonlinear leader-following multi-agent systems with input-delay [27]. In order to reduce transmission times of packets, Zhang et al. [28] introduced an event-based packet processor on the basis of sampling measurement output. To eliminate the Zeno phenomenon compulsively, each triggering instant is performed only after a preset time interval [29]. In addition to static triggering approach designed for aforementioned control strategies, the dynamic triggering approach has been proposed as an effective method to enlarge the inter-event time [30]. Hu et al. [31] set event-triggered rules based on a dynamic function to reach stability of multi-agent systems under directed networks. He et al. [32] proposed a distributed dynamic triggering mechanism with a dynamic threshold. By introducing an additional parameter, the threshold used to determine triggering instants is calculated dynamically. Luo et al. [33] designed two triggering conditions separately to intermittently broadcast information among neighbors and reduce controller updates.

Although the dynamic triggering mechanisms mentioned above have made significant contributions to reducing communication costs, some scalars related to the global information of network topology are still needed in designing controllers or triggering conditions. This disadvantage poses many difficulties especially for large-scale networks. Therefore, many scholars have aimed to develop fully distributed event-triggered controllers and triggering functions. Some of these efforts include the work in [34–36]. Cheng et al. [34] introduced the edge-based adaptive parameters to realize the consensus of multi-agent systems without global system information. Liu et al. [35] proposed a novel dynamic double-event-triggered mechanism and designed a controller with the node-based adaptive parameter to reach the fully distributed consensus of multi-agent systems. Li et al. [36] designed fully distributed event/self-triggered mechanism for tracking control of heterogeneous multi-agent systems. However, their proposed algorithms avoided Zeno problem by proving that the length of triggering interval is always greater than zero, which is not enough to ensure their practical implementation. This is because an inter-event time that indefinitely approaches zero can make it difficult for physical hardware to meet the required response speed of the event-triggered algorithm. In addition, such Zeno-freeness can be easily destroyed by external disturbances. By contrast, if it is proved that the length of triggering interval is always larger than a positive constant, this algorithm can be considered to have strong anti-interference ability and practicability. In 2020, Berneburg et al. [37] realized this problem and derived exact lower bounds of inter-event time to prevent Zeno phenomenon, but this approach was only designed for single-integrator multi-agent systems without leaders. Recently, Qian et al. [38] developed a dynamic adaptive event-triggered algorithm in terms of leaderless consensus of multi-agent systems, which is fully distributed and can also obtain exact lower bounds of inter-event time. Although adaptive parameters are set for each edge in the topological graph, the controller and triggering condition are till designed for each node, and the sum of relative information of all neighbors are still needed. Moreover, for the connected large-scale graph, compared with the setting of adaptive parameters corresponding to nodes, the setting of adaptive parameters corresponding to edges increases the computation and occupies more memory space. In 2022, Wu et al. [39] extended the triggering mechanism proposed in [37] and presented a dynamic event-triggered algorithm for tracking control of multi-agent systems. However, their triggering function and feedback gain matrix rely on the global features of system networks, that is the algorithm is not fully distributed.

Inspired by the aforementioned studies, this paper takes a step forward toward designing a flexible and strong anti-jamming adaptive event-triggered strategy for leader-following and leaderless multi-agent systems that does not use global features of system networks. The contributions of this note can be summarized in the following four aspects:

- A flexible event-triggered mechanism with a virtual internal variable is designed, which not only reduces communication energy consumption, but also improves anti-interference ability and practicability by deducing the minimum length of triggering intervals for each agent. These advantages improve the feasibility of it in practical applications compared with works in [30–36].
- A fully distributed event-triggered controller using adaptive node-based parameter is designed for the multi-agent system with a leader, which utilizes the estimated state values of agents based on actual information at triggering instants. In this way, global information is denied on the basis of reducing the communication burdens and controller update frequency, which is impossible for approaches in [30–33,39].
- Convergence conditions independent of global network information are obtained for the multi-agent systems via Lyapunov stability theory. These conditions are less conservative compared to those obtained in [32,39].
- The convergence conditions for leaderless multi-agent systems based on the designed fully distributed dynamic event-triggered mechanism are further derived.

Notation: $\mathbb{N}$ and $\mathbb{R}$ separately represent a set of nonnegative integers and a set of real numbers. $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$ are n dimensional vector space and $m\times n$ real matrix space. I_n and $\mathbf{0}$ separately stand for the $n\times n$ identity matrix and the all-zero column vector. $|\cdot|$ represents the module of a matrix. The transpose and eigenvalues of matrix D can be represented as D^T and $\lambda(D)$, respectively. $z(t^-)$ and $z(t^+)$ are the left and right limits of $z(t)$, respectively. For simplicity, we write z to stand for $z(t)$ if it can be understood from context. The maximum and minimum value of z can be defined as $z_{\max}$ and $z_{\min}$, respectively.

2. Preliminaries and problem expression

2.1. Graph theory

The communication links among agents in the leader-following system can be showed with a graph $\tilde{G}$. Suppose there exist M followers in the system, and the topological structure of them is described with the undirected graph $G = (\mathbb{V}, \mathbb{E}, \mathbb{A})$. G is a subgraph of $\tilde{G}$. $\mathbb{V} = \{v_1, v_2, \dots, v_M\}$ represents the node set, $\mathbb{E} = \{\epsilon_{ij} = (v_i, v_j)\}$ represents the edge set and $\mathbb{A} = [\alpha_{ij}] \in \mathbb{R}^{M\times M}$ represents the adjacency matrix. Particularly, $\alpha_{ij} = 1$ if $\epsilon_{ij} \in \mathbb{E}$; otherwise, $\alpha_{ij} = 0$. The neighbor of i can be represented as $R_i = \{v_j \in \mathbb{V} | \alpha_{ij} = 1\}$. Each node in G can connect with any other nodes if and only if G is connected. $L \in \mathbb{R}^{M\times M}$ is the corresponding Laplacian matrix in which each diagonal element is equal to $\sum_{j\in R_i} \alpha_{ij}$ and each non-diagonal element is equal to $-\alpha_{ij}$. In addition, there is at least one edge in $\tilde{G}$ that points from the leader to a follower. The connectivity from the leader to followers can be expressed with $C = \mathrm{diag}\{c_1, c_2, \dots, c_M\}$. $c_i = 1$ if there is an edge that points to ith follower from the leader, otherwise, $c_i = 0$.

2.2. Problem expression

The follower and the leader in $\tilde{G}$ fulfill the following general linear dynamic

$$\begin{aligned} &\dot{z}_i(t) = Dz_i(t) + Eu_i(t), i = 1, 2, \dots, M, \\ &\dot{z}_0(t) = Dz_0(t) \end{aligned} \tag{1}$$

in which $z_i(t) \in \mathbb{R}^n$ and $u_i(t) \in \mathbb{R}^m$ are state vector and control input vector corresponding to followers. $z_0(t) \in \mathbb{R}^n$ is the state vector corresponding to the leader. $D \in \mathbb{R}^{n\times n}$ and $E \in \mathbb{R}^{n\times m}$ are system matrices.

Definition 1. Given $z_w(0)$, $w = 0, 1, 2, \ldots, M$ as initial states of agents, if

$$\lim_{t\to\infty} \|z_i(t) - z_0(t)\| = 0, i = 1, 2, \ldots, M, \tag{2}$$

the consensus of (1) is realized for the leader-following multi-agent systems.

To achieve the consensus of (1), a fully distributed event-triggered approach following adaptive control rules will be designed. The following assumptions and lemma will contribute to subsequent analysis.

Assumption 1. (D, E) is stabilizable.

Assumption 2. The undirected graph G is connected.

Lemma 1 ([40]). *For the algebraic Riccati equation (ARE)*

$$D^T P + PD - PEE^T P + F = 0 \tag{3}$$

where $F = F^T > 0$, the symmetric matrix P is positive definite if (D, E) is stabilizable.

3. Main results

A new event-triggered control rule will be constructed in this part, and the stability of (1) with it will be analyzed.

3.1. Adaptive node-based controller design

Generally, each follower can get information from its neighboring agents. The relative information for ith follower can be integrated as

$$\xi_i = \sum_{j\in R_i} \alpha_{ij}(z_j - z_i) + c_i(z_0 - z_i), i = 1, 2, \ldots, M. \tag{4}$$

Considering real-time state information of agents are avoided for event-triggered control, the corresponding form of ξ_i is represented as

$$\hat{\xi}_i = \sum_{j\in R_i} \alpha_{ij}(\hat{z}_j - \hat{z}_i) + c_i(\hat{z}_0 - \hat{z}_i), i = 1, 2, \ldots, M \tag{5}$$

in which $\hat{z}_w, w = 0, 1, 2, \ldots, M$ denotes the estimated variable of z_w. The assignment rule of it is

$$\begin{cases} \hat{z}_w(t) = z_w(t), & t = t_k^w, \\ \dot{\hat{z}}_w(t) = D\hat{z}_w(t), & t \in (t_k^w, t_{k+1}^w). \end{cases} \tag{6}$$

Obviously, $\hat{z}_0(t) = z_0(t)$.

For the sake of avoiding the real-time relative information of each follower and flexibly adjusting the control input in accordance with convergence process, the controller of ith follower is designed as

$$u_i = J\chi_i\hat{\xi}_i, \tag{7}$$

in which $J = E^T P$. P is the matrix satisfying ARE (3). χ_i is called adaptive control parameter satisfying

$$\dot{\chi}_i = \begin{cases} 0, & \chi_i \ge \bar{\chi}_i, \\ \kappa_i\hat{\xi}_i^T \Omega\hat{\xi}_i, & \chi_i < \bar{\chi}_i, \end{cases} \tag{8}$$

where κ_i, $\bar{\chi}_i > 0$, and $\Omega = PEE^T P$. Given the initial value $0 < \chi_i(0) < \bar{\chi}_i$, the updating rule (8) not only guarantees that χ_i is obtained according to the variation in convergence performance, but also guarantees to limit it within a reasonable range.

Remark 1. Note that the adaptive event-triggered controller (7) needs the estimated state values of agents, which can be calculated by (6). Obviously, the real-time state information of agents is required only at the triggering instants, which means the continuous state monitoring and transmission are no longer necessary. In this way, the communication burdens of the system can be greatly reduced.

Remark 2. Compared with the controllers designed in [30–33,39], the feedback gain matrix J is obtained by solving the ARE (3) without any global features of the system network. χ_i is also got without the network scale and any eigenvalues of the Laplacian matrix. Therefore, the controller designed in this paper can work in a fully distributed criteria. In addition, compared with edge-based adaptive parameters designed in [38,41], the node-based ones designed in (8) can greatly reduce the computation and the occupation of memory space especially for the connected large-scale graph. Moreover, by adjusting $\bar{\chi}_i$, the range of χ_i can be flexibly adjusted.

3.2. Dynamic event-triggered mechanism

Considering the significant advantages of dynamic event-triggered strategy in reducing triggering frequency and increasing the length of triggering interval, we will design a dynamic triggering mechanism determined by a virtual internal variables in this section.

Set $\{t_k^i : k \in \mathbb{N}, i = 1, 2, \ldots, M\}$ as the triggering time sequence, and $t_0^i = 0$. To design the dynamic event-triggered strategy, a virtual internal variable $\psi_i(t)$ corresponding to each follower is introduced. The derivative of ψ_i is given by

$$\dot{\psi}_i = \begin{cases} -\beta_i, & \|\varepsilon_i\| = 0, \\ -[\chi_i^2\psi_i^2 + \psi_i + \phi_i]\dfrac{\varepsilon_i^T\Omega\varepsilon_i}{\varepsilon_i^T P\varepsilon_i} - \beta_i, & \|\varepsilon_i\| \neq 0, \end{cases} \tag{9}$$

where $\varepsilon_i = \hat{z}_i - z_i$ is the estimate error of ith follower, and β_i is a positive constant which can be determined in advance. $\phi_i(t)$ is an adaptive parameter which satisfies $\dot{\phi}_i = \sigma_i\varepsilon_i^T\Omega\varepsilon_i$, $\sigma_i > 0$ and $\phi_i(0) > 0$. Obviously, $\dot{\psi}_i < 0$.

Let $\psi_i(0) = \bar{\psi}_i$ where $\bar{\psi}_i > 0$ is a constant which can be determined in advance. At any given time $t > t_k^i, k \in \mathbb{N}$, the next triggering instant for ith follower is considered as

$$t_{k+1}^i = \min\{t > t_k^i | \psi_i(t) \le 0\}. \tag{10}$$

Once the new event is triggered, the internal variable ψ_i is reset to $\bar{\psi}_i$.

Combining with the dynamics of ψ_i, one can clearly analyze the changing law of it. When an event is triggered, ε_i is set to 0 at once. $\dot{\psi}_i = -\beta_i < 0$ makes the variable ψ_i start to decrease. When $t_k^i < t < t_{k+1}^i$, $\dot{\psi}_i < 0$. The variable ψ_i continues to decrease. When it reduces to 0, the triggering condition is satisfied, and ψ_i is reset to $\bar{\psi}_i$. Therefore, ψ_i is decreased when $t \in [t_k^i, t_{k+1}^i), \forall k \in \mathbb{N}$, and $0 \le \psi_i \le \bar{\psi}_i$ for $\forall t \ge 0$.

Remark 3. A dynamic event-triggered approach with a virtual internal variable is introduced, which makes it possible for each agent to communicate with its neighbors intermittently. The triggering mechanism (10) guarantees that each agent can trigger events asynchronously. Compared with the event-triggered strategies with dynamic threshold proposed in [32,38,39], the mechanism (10) can work without any relative real-time or triggering instants information of neighboring agents, and global information of system networks is also not necessary. In addition, the proposed mechanism is more concise in construction and more explicit in operation. The triggering instants can be easily adjusted by changing $\bar{\psi}_i$, $\bar{\chi}_i$, $\phi_i(0)$ and σ_i. Furthermore, as will be shown in Theorem 2, a precise positive minimum length of triggering intervals for each follower can be obtained with the proposed triggering approach, which can not only eliminate the Zeno phenomenon, but also improve the anti-interference ability and practical operability of the algorithm. These advantages can improve the feasibility of it in practical application.

Under the dynamic triggering approach (10), the adaptive dynamic event-triggered leader-following consensus algorithm for multi-agent systems with a leader using the controller (7) can be presented as Algorithm 1.

Algorithm 1 Adaptive dynamic event-triggered leader-following consensus algorithm

Input: initial variables $z_0(0), z_i(0), \chi_i(0), \bar{\chi}_i, , \psi_i(0), \phi_i(0)$ and $\beta_i > 0$, $i = 1, 2, \cdots, M$;
Output: $z_0(t)$, $z_i(t)$, $\chi_i(t)$, $i = 1, 2, \cdots, M$;
Update z_0 according to (1);
for $i = 1, 2, \dots, M$ **do**
  if $\psi_i <= 0$ **then**
    Update $\hat{z}_i$ to z_i and broadcast the state information to its neighbors;
    Reset the clock timer $\psi_i = \bar{\psi}_i$;
  else
    Update ψ_i according to the dynamics (9);
    Update $\hat{z}_i$ according to the dynamics (6);
  end if
  Update χ_i according to (8);
  Update the event-triggered controller (7);
  Update state information of ith follower according to (1);
  Update ϕ_i;
end for

3.3. Leader-following consensus analysis

Considering the adaptive control protocol (7), the derivative of each follower can be rewritten as

$$\dot{z}_i = Dz_i + EJ\chi_i\hat{\xi}_i. \tag{11}$$

The compact matrix form is

$$\dot{z} = (I_M \otimes D)z + (\chi \otimes EJ)\hat{\xi}, \tag{12}$$

where $z = [z_1^T, z_2^T, \dots, z_M^T]^T$, $\chi = \operatorname{diag}\{\chi_1, \chi_2, \dots, \chi_M\}$ and $\hat{\xi} = [\hat{\xi}_1^T, \hat{\xi}_2^T, \dots, \hat{\xi}_M^T]^T$.

Let $\varsigma_i = z_i - z_0$. Then, one gets

$$\dot{\varsigma} = (I_M \otimes D)z + (\chi \otimes EJ)\hat{\xi}. \tag{13}$$

where $\varsigma = [\varsigma_1^T, \varsigma_2^T, \dots, \varsigma_M^T]^T$ is the consensus error. The tracking consensus of system (1) is reached if and only if $\lim_{t\to\infty} \varsigma = \mathbf{0}$.

Theorem 1. *If the multi-agent system (1) satisfies Assumptions 1 and 2, the fully distributed adaptive controller (7) with the dynamic event-triggered protocol (10) is able to solve the tracking consensus asymptotically when* $\bar{\chi}_i \geq 1$.

Proof. Define the following Lyapunov function

$$V = \varsigma^T(H \otimes P)\varsigma + \sum_{i=1}^{M}\left[\frac{(\chi_i - 1 - a)^2}{2\kappa_i} + \frac{(\phi_i - b)^2}{2\sigma_i} + \psi_i\varepsilon_i^T P\varepsilon_i\right] \tag{14}$$

where $H = L + C$. It is easy to get $V \geq 0$. For convenience, we set $V_1 = \varsigma^T(H \otimes P)\varsigma$, and $V_2 = \sum_{i=1}^{M}\left[\frac{(\chi_i - 1 - a)^2}{2\kappa_i} + \frac{(\phi_i - b)^2}{2\sigma_i} + \psi_i\varepsilon_i^T P\varepsilon_i\right]$. Then, take the time derivatives of V_1 and V_2 for $t \in (t_k, t_{k+1}), \forall k \in \mathbb{N}$.

Firstly, the derivative of V_1 can be obtained as follows:

$$\dot{V}_1 = 2\varsigma^T(H \otimes PD)\varsigma + 2\varsigma^T(H\chi \otimes \Omega)\hat{\xi}. \tag{15}$$

Let $\xi = [\xi_1^T, \xi_2^T, \dots \xi_M^T]^T$, $\eta_i = \sum_{j\in R_i}\alpha_{ij}(\varepsilon_j - \varepsilon_i) - c_i\varepsilon_i = \hat{\xi}_i - \xi_i$, and $\eta = [\eta_1^T, \eta_2^T, \dots \eta_M^T]^T$. One gets $\xi = -(H \otimes I_n)\varsigma$ and $\eta = \hat{\xi} - \xi$. Thus,

$$\begin{aligned}\dot{V}_1 &= 2\varsigma^T(t)(H \otimes PD)\varsigma - 2\xi^T(\chi \otimes \Omega)\hat{\xi}\\ &= 2\varsigma^T(H \otimes PD)\varsigma - 2\hat{\xi}^T(\chi \otimes \Omega)\hat{\xi} + 2\eta^T(\chi \otimes \Omega)\hat{\xi}\\ &\leq 2\varsigma^T(H \otimes PD)\varsigma - \hat{\xi}^T(\chi \otimes \Omega)\hat{\xi} + \eta^T(\chi \otimes \Omega)\eta.\end{aligned} \tag{16}$$

Secondly, take the time derivative of V_2,

$$\begin{aligned}\dot{V}_2 = {}& \sum_{i=1}^{M}\left[\frac{(\chi_i - 1 - a)\dot{\chi}_i}{\kappa_i} + \frac{(\phi_i - b)\dot{\phi}_i}{\sigma_i}\right] + \sum_{i=1}^{M}\dot{\psi}_i\varepsilon_i^T P\varepsilon_i +\\ & 2\sum_{i=1}^{M}\psi_i\varepsilon_i^T P\dot{\varepsilon}_i\\ = {}& \sum_{i=1}^{M}\left[\frac{(\chi_i - 1 - a)\dot{\chi}_i}{\kappa_i} + (\phi_i - b)\varepsilon_i^T\Omega\varepsilon_i\right] + \sum_{i=1}^{M}\dot{\psi}_i\varepsilon_i^T P\varepsilon_i +\\ & 2\sum_{i=1}^{M}\psi_i\varepsilon_i^T PD\varepsilon_i - 2\sum_{i=1}^{M}\chi_i\psi_i\varepsilon_i^T\Omega\hat{\xi}_i.\end{aligned} \tag{17}$$

According to the Young's inequality $ab \leq \frac{1}{2h}a^2 + \frac{h}{2}b^2, \forall a, b \in R, h > 0$, one has

$$-2\sum_{i=1}^{M}\chi_i\psi_i\varepsilon_i^T\Omega\hat{\xi}_i \leq \sum_{i=1}^{M}\chi_i^2\psi_i^2\varepsilon_i^T\Omega\varepsilon_i + \sum_{i=1}^{M}\hat{\xi}_i^T\Omega\hat{\xi}_i. \tag{18}$$

According to Lemma 1,

$$\begin{aligned}2\sum_{i=1}^{M}\psi_i\varepsilon_i^T PD\varepsilon_i &= \sum_{i=1}^{M}\psi_i\varepsilon_i^T(D^T P + PD - \Omega)\varepsilon_i + \sum_{i=1}^{M}\psi_i\varepsilon_i^T\Omega\varepsilon_i\\ &\leq \sum_{i=1}^{M}\psi_i\varepsilon_i^T\Omega\varepsilon_i.\end{aligned} \tag{19}$$

Therefore,

$$\begin{aligned}\dot{V} \leq {}& 2\varsigma^T(H \otimes PD)\varsigma + \sum_{i=1}^{M}\frac{(\chi_i - 1 - a)\dot{\chi}_i}{\kappa_i} - \sum_{i=1}^{M}b\varepsilon_i^T\Omega\varepsilon_i +\\ & \sum_{i=1}^{M}(1 - \chi_i)\hat{\xi}_i^T\Omega\hat{\xi}_i + \sum_{i=1}^{M}\chi_i\eta_i^T\Omega\eta_i + \sum_{i=1}^{M}\dot{\psi}_i\varepsilon_i^T P\varepsilon_i +\\ & \sum_{i=1}^{M}\psi_i\varepsilon_i^T\Omega\varepsilon_i + \sum_{i=1}^{M}\chi_i^2\psi_i^2\varepsilon_i^T\Omega\varepsilon_i + \sum_{i=1}^{M}\phi_i\varepsilon_i^T\Omega\varepsilon_i.\end{aligned} \tag{20}$$

Based on the piecewise derivative of ψ_i as shown in (9), one can continue to analyze $\dot{V}$ in two cases.

(1) When $\|\varepsilon_i\| = 0$, it is obvious that

$$\begin{aligned}\dot{V} \leq {}& 2\varsigma^T(H \otimes PD)\varsigma + \sum_{i=1}^{M}\frac{(\chi_i - 1 - a)\dot{\chi}_i}{\kappa_i} + \sum_{i=1}^{M}(1 - \chi_i)\hat{\xi}_i^T\Omega\hat{\xi}_i +\\ & \sum_{i=1}^{M}\chi_i\eta_i^T\Omega\eta_i.\end{aligned} \tag{21}$$

If $\chi_i < \bar{\chi}_i$, one can simplify $\dot{V}$ to

$$\dot{V} \leq 2\varsigma^T(H \otimes PD)\varsigma - \sum_{i=1}^{M}a\hat{\xi}_i^T\Omega\hat{\xi}_i + \sum_{i=1}^{M}\chi_i\eta_i^T\Omega\eta_i. \tag{22}$$

Based on $\hat{\xi}_i = \xi_i + \eta_i$ and Young's inequality, one gets

$$-\sum_{i=1}^{M}a\hat{\xi}_i^T\Omega\hat{\xi}_i \leq -\sum_{i=1}^{M}a\left(1 - \frac{1}{h}\right)\xi_i^T\Omega\xi_i + \sum_{i=1}^{M}a(h - 1)\eta_i^T\Omega\eta_i \tag{23}$$

where $h > 1$. Then,

$$\begin{aligned}&\sum_{i=1}^{M}a(h - 1)\eta_i^T\Omega\eta_i = \sum_{i=1}^{M}a(h - 1)\left\|\eta_i^T PE\right\|^2\\ &\leq \sum_{i=1}^{M}a(h - 1)\Big(H_{ii}\left\|\varepsilon_i^T PE\right\| + \sum_{j\in R_i}\alpha_{ij}\left\|\varepsilon_j^T PE\right\|\Big)^2\\ &\leq \sum_{i=1}^{M}a(h - 1)\mathcal{M}_i\varepsilon_i^T\Omega\varepsilon_i\end{aligned} \tag{24}$$

where $\mathcal{M}_i = (|R_i| + 1)(H_{ii}^2 + \sum_{j\in R_i}\alpha_{ji}^2)$. Similarly,

$$\sum_{i=1}^{M}\chi_i\eta_i^T\Omega\eta_i \leq \sum_{i=1}^{M}\mathcal{N}_i\varepsilon_i^T\Omega\varepsilon_i \tag{25}$$

where $\mathcal{N}_i = (|R_i| + 1)(H_{ii}^2\bar{\chi}_i + \sum_{j\in R_i}\alpha_{ji}^2\bar{\chi}_j)$. Thus,

$$\dot{V} \leq 2\varsigma^T(H \otimes PD)\varsigma - \sum_{i=1}^{M}a\left(1 - \frac{1}{h}\right)\xi_i^T\Omega\xi_i. \tag{26}$$

Under Assumption 2, there exists an orthogonal matrix U such that $U^T H U = \Lambda = diag\{\lambda_1(H), \lambda_2(H), \ldots, \lambda_M(H)\}$, and $\lambda_i(H) > 0$ for $i = 1, 2, \ldots, M$. Let

$$\breve{\zeta} = [\breve{\zeta}_1^T, \breve{\zeta}_2^T, \ldots, \breve{\zeta}_M^T]^T = (U^T \otimes I_n)\varsigma. \tag{27}$$

On gets $\varsigma = (U \otimes I_n)\breve{\zeta}$. Then, considering $P = P^T > 0$ one obtains

$$\begin{aligned} 2\varsigma^T (H \otimes PD)\varsigma &= 2\breve{\zeta}^T (U^T \otimes I_n)(H \otimes PD)(U \otimes I_n)\breve{\zeta} \\ &= \sum_{i=1}^{M} \lambda_i(H)\breve{\zeta}_i^T (D^T P + PD)\breve{\zeta}_i. \end{aligned} \tag{28}$$

Supposing $a \geq \frac{h}{(h-1)\lambda_{\min}(H)} > 0$, one obtains

$$\sum_{i=1}^{M} a(1 - \frac{1}{h})\xi_i^T \Omega \xi_i \geq \sum_{i=1}^{M} \frac{1}{\lambda_{\min}(H)} \xi_i^T \Omega \xi_i \geq \sum_{i=1}^{M} \frac{1}{\lambda_i(H)} \xi_i^T \Omega \xi_i. \tag{29}$$

Since $\xi = -(H \otimes I_n)\varsigma$ and $\varsigma = (U \otimes I_n)\breve{\zeta}$, one gets

$$\begin{aligned} \sum_{i=1}^{M} \xi_i^T \Omega \xi_i &= \xi^T (I_M \otimes \Omega)\xi \\ &= ((HU \otimes I_n)\breve{\zeta})^T (I_M \otimes \Omega)((HU \otimes I_n)\breve{\zeta}) \\ &= \breve{\zeta}^T (U^T H^T U U^T H U \otimes \Omega)\breve{\zeta} \\ &= \breve{\zeta}^T (\Lambda^2 \otimes \Omega)\breve{\zeta}. \end{aligned} \tag{30}$$

Therefore,

$$\sum_{i=1}^{M} a(1 - \frac{1}{h})\xi_i^T \Omega \xi_i \geq \sum_{i=1}^{M} \lambda_i(H)\breve{\zeta}_i^T \Omega \breve{\zeta}_i. \tag{31}$$

(26) is rewritten as

$$\dot{V} \leq \sum_{i=1}^{M} \lambda_i(H)\breve{\zeta}_i^T (D^T P + PD - \Omega)\breve{\zeta}_i \leq -\sum_{i=1}^{M} \lambda_i(H)\breve{\zeta}_i^T F \breve{\zeta}_i \leq 0. \tag{32}$$

If $\chi_i \geq \bar{\chi}_i$, $\dot{\chi}_i = 0$ which means $\chi_i = \bar{\chi}_i \geq 1$. (21) can be simplified to

$$\dot{V} \leq 2\varsigma^T (H \otimes PD)\varsigma + \sum_{i=1}^{M} (1 - \bar{\chi}_i)\hat{\xi}_i^T \Omega \hat{\xi}_i + \sum_{i=1}^{M} \bar{\chi}_i \eta_i^T \Omega \eta_i. \tag{33}$$

It is similar to the derivation from (22) to (26), and one obtains

$$\dot{V} \leq 2\varsigma^T (H \otimes PD)\varsigma + \sum_{i=1}^{M} (1 - \bar{\chi}_i)(1 - \frac{1}{r})\xi_i^T \Omega \xi_i \tag{34}$$

where $r > 1$. Let $r > 1 + \frac{1}{\lambda_{\min}(H)(\tilde{\chi} - 1) - 1}$ where $\tilde{\chi} = \min\{\bar{\chi}_i, \forall i = 1, 2, \cdots, M\}$. One gets $\dot{V} \leq 0$ with the similar derivation from (26) to (32).

(2) When $\|\varepsilon_i\| \neq 0$, $\dot{\psi}_i = -[\chi_i^2 \psi_i^2 + \psi_i + \phi_i]\frac{\varepsilon_i^T \Omega \varepsilon_i}{\varepsilon_i^T P \varepsilon_i} - \beta_i$. Then,

$$\begin{aligned} \dot{V} \leq & 2\varsigma^T (H \otimes PD)\varsigma + \sum_{i=1}^{M} \frac{(\chi_i - 1 - a)\dot{\chi}_i}{\kappa_i} - \sum_{i=1}^{M} b\varepsilon_i^T \Omega \varepsilon_i + \\ & \sum_{i=1}^{M} (1 - \chi_i)\hat{\xi}_i^T \Omega \hat{\xi}_i + \sum_{i=1}^{M} \chi_i \eta_i^T \Omega \eta_i. \end{aligned} \tag{35}$$

If $\chi_i < \bar{\chi}_i$, one can simplify $\dot{V}$ to

$$\dot{V} \leq 2\varsigma^T (H \otimes PD)\varsigma - \sum_{i=1}^{M} a\hat{\xi}_i^T \Omega \hat{\xi}_i - \sum_{i=1}^{M} b\varepsilon_i^T \Omega \varepsilon_i + \sum_{i=1}^{M} \chi_i \eta_i^T \Omega \eta_i. \tag{36}$$

Combining with $\|\varepsilon_i\| \neq 0$, by the similar derivation from (22) to (26), one gets

$$\begin{aligned} \dot{V} \leq & 2\varsigma^T (H \otimes PD)\varsigma - \sum_{i=1}^{M} a(1 - \frac{1}{p})\xi_i^T \Omega \xi_i + \\ & \sum_{i=1}^{M} [\mathcal{M}_i a(p-1) + \mathcal{N}_i - b]\varepsilon_i^T \Omega \varepsilon_i \end{aligned} \tag{37}$$

where $p > 1$. Let $b \geq \tilde{\mathcal{M}} a(p-1) + \tilde{\mathcal{N}}$ where $\tilde{\mathcal{M}} = \max\{\mathcal{M}_i, \forall i = 1, 2, \ldots, M\}$ and $\tilde{\mathcal{N}} = \max\{\mathcal{N}_i, \forall i = 1, 2, \ldots, M\}$. One obtains

$$\dot{V} \leq 2\varsigma^T (H \otimes PD)\varsigma - \sum_{i=1}^{M} a(1 - \frac{1}{p})\xi_i^T \Omega \xi_i. \tag{38}$$

Supposing $a \geq \frac{p}{(p-1)\lambda_{\min}(H)}$ and combining with Assumption 2, (38) is rewritten as

$$\dot{V} \leq \sum_{i=1}^{M} \lambda_i(H)\breve{\zeta}_i^T (D^T P + PD - \Omega)\breve{\zeta}_i \leq -\sum_{i=1}^{M} \lambda_i(H)\breve{\zeta}_i^T F \breve{\zeta}_i \leq 0. \tag{39}$$

If $\chi_i \geq \bar{\chi}_i$, (35) can be simplified to

$$\dot{V} \leq 2\varsigma^T (H \otimes PD)\varsigma + \sum_{i=1}^{M} (1 - \bar{\chi}_i)\hat{\xi}_i^T \Omega \hat{\xi}_i - \sum_{i=1}^{M} b\varepsilon_i^T \Omega \varepsilon_i + \sum_{i=1}^{M} \bar{\chi}_i \eta_i^T \Omega \eta_i. \tag{40}$$

Similarly,

$$\dot{V} \leq 2\varsigma^T (H \otimes PD)\varsigma + \sum_{i=1}^{M} (1 - \bar{\chi}_i)(1 - \frac{1}{q})\xi_i^T \Omega \xi_i + \sum_{i=1}^{M} (q\mathcal{N}_i - b)\varepsilon_i^T \Omega \varepsilon_i \tag{41}$$

where $q > 1$. Let $b \geq q\mathcal{N}_i$ and $q > 1 + \frac{1}{\lambda_{\min}(H)(\tilde{\chi} - 1) - 1}$. Considering Assumption 2 and ARE (3), one gets $\dot{V} \leq 0$.

To sum up, it shows that $\dot{V}(t)$ is non-positive for $t \in (t_k, t_{k+1}), \forall k \in \mathbb{N}$. In addition, considering $\psi_i(t_k^-) = 0$, $\psi_i(t_k^+) = \bar{\psi}_i$ and $\varepsilon_i(t_k^+) = 0$, V does not change at the triggering instants. Therefore, $\dot{V}(t)$ is non-positive over $[t_k, t_{k+1}], \forall k \in \mathbb{N}$. Recalling that $V(t)$ is non-negative, it is inferred that $V(t)$ is bounded over $t \in [0, \infty)$. $\chi_i(t)$ and $\phi_i(t)$ can approach to $\bar{\chi}_i$ and $\bar{\phi}_i$ as $t \to \infty$. Following LaSalle's invariance principle, one gets $\lim_{t\to\infty} \varsigma(t) = 0$. Thus, one can conclude that the tracking consensus of system (1) is achieved asymptotically. The proof is completed.

Remark 4. One can notice that the variable ψ_i always jumps from 0 to $\bar{\psi}_i$ once the event is triggered, i.e. $\psi_i(t_k^-) = 0$ and $\psi_i(t_k^+) = \bar{\psi}_i$. The last term of (14) can avoid the effect on V caused by this jump.

Subsequently, we will prove that the proposed event-triggered strategy (10) can remove Zeno phenomenon for the multi-agent system with a leader.

Theorem 2. *For the multi-agent system (1) with the proposed triggering approach, Zeno phenomenon can be removed by getting a positive minimum length of triggering intervals for each follower when the condition derived in Theorem 1 is satisfied. The minimum length of triggering intervals for each follower is*

$$\tau_i = \frac{2}{\tilde{\Theta}}(\arctan \frac{2\Theta_1 \bar{\psi}_i + \Theta_2}{\tilde{\Theta}} - \arctan \frac{\Theta_2}{\tilde{\Theta}}) > 0 \tag{42}$$

where $\tilde{\Theta} = \sqrt{4\Theta_1\Theta_3 - \Theta_2^2}$, $\Theta_1 = \frac{\bar{\chi}_i^2 \lambda_{\max}(\Omega)}{\lambda_{\min}(P)}$, $\Theta_2 = \frac{\lambda_{\max}(\Omega)}{\lambda_{\min}(P)}$ *and* $\Theta_3 = \frac{\bar{\phi}_i \lambda_{\max}(\Omega)}{\lambda_{\min}(P)} + \beta_i$.

Proof. According to the proposed triggering approach, we can infer that the inter-event time for ith follower is equivalent to the time interval when the ψ_i returns to 0 from $\bar{\psi}_i$. Thus, we need to figure out the inter-event time with the derivative of ψ_i as shown in (9). The basic strategy is to get the lower bound of $t_{k+1}^i - t_k^i$ by using the integral of lower bound for $\dot{\psi}_i$, $\psi_i(t_k^{i+}) = \bar{\psi}_i$ and $\psi_i(t_{k+1}^{i-}) = 0$. Obviously,

$$\dot{\psi}_i \geq -[\chi_i^2 \psi_i^2 + \psi_i + \phi_i]\frac{\varepsilon_i^T \Omega \varepsilon_i}{\varepsilon_i^T P \varepsilon_i} - \beta_i. \tag{43}$$

Considering the initial variables and updating laws of χ_i and ϕ_i, they can converge to $\bar{\chi}_i$ and $\bar{\phi}_i$ as $t \to \infty$, one gets

$$\dot{\psi}_i \geq -[\bar{\chi}_i^2 \psi_i^2 + \psi_i + \bar{\phi}_i]\frac{\lambda_{\max}(\Omega)}{\lambda_{\min}(P)}. \tag{44}$$

Define

$$\dot{\vartheta}_i = -\Theta_1\vartheta_i^2 - \Theta_2\vartheta_i - \Theta_3 \tag{45}$$

with $\Theta_1 = \frac{\bar{\chi}_i^2\lambda_{\max}(\Omega)}{\lambda_{\min}(P)}$, $\Theta_2 = \frac{\lambda_{\max}(\Omega)}{\lambda_{\min}(P)}$, $\Theta_3 = \frac{\bar{\phi}_i\lambda_{\max}(\Omega)}{\lambda_{\min}(P)} + \beta_i$, $\vartheta_i(t_k^{i+}) = \bar{\psi}_i$ and $\vartheta_i(t_{k+1}^{i-}) = 0$. By taking the integral of (45) on $[t_k^i, t_{k+1}^i)$, it can be obtained that

$$t_{k+1}^i - t_k^i = \frac{2}{\tilde{\Theta}}(\arctan\frac{2\Theta_1\bar{\psi}_i + \Theta_2}{\tilde{\Theta}} - \arctan\frac{\Theta_2}{\tilde{\Theta}}) > 0. \tag{46}$$

According to the Lemma 3.4 in [42], it follows that $\psi_i \geq \vartheta_i$. Therefore, the minimum length of triggering intervals for ith follower is

$$\tau_i = \frac{2}{\tilde{\Theta}}(\arctan\frac{2\Theta_1\bar{\psi}_i + \Theta_2}{\tilde{\Theta}} - \arctan\frac{\Theta_2}{\tilde{\Theta}}) > 0. \tag{47}$$

Zeno behavior is eliminated. The proof is completed.

Remark 5. As is known to all, in order to measure the feasibility of triggering algorithms, theoretical analysis to break Zeno phenomenon is essential. Numerous existing triggering approaches usually break Zeno phenomenon by inferring that the triggering interval is always greater than zero. However, one should notice that too small positive inter-event time which indefinitely approaches zero will make it difficult for physical hardware to meet the required response speed of the event-triggered algorithm. In addition, such Zeno-freeness can be easily destroyed by external disturbances. By contrast, if it is proved that the length of triggering interval is always larger than a positive constant, this algorithm can be considered to have strong anti-interference ability and practicability. From this point of view, the algorithm in this paper has greater advantages than others in [31–36]. In addition, the length of triggering interval can be adjusted by changing $\bar{\psi}_i$, $\bar{\chi}_i$, $\phi_i(0)$ and β_i.

Next, the consensus of leaderless multi-agent systems based on event-triggered approach is addressed. The dynamic of ith agent follows

$$\dot{z}_i = Dz_i + Eu_i, i = 1, 2, \dots, M, \tag{48}$$

in which u_i satisfies

$$u_i = J\varrho_i\hat{\zeta}_i. \tag{49}$$

$\hat{\zeta}_i = \sum_{j\in R_i}\alpha_{ij}(\hat{z}_j - \hat{z}_i)$. ϱ_i is an adaptive parameter satisfying

$$\dot{\varrho}_i = \begin{cases} 0, & \varrho_i \geq \bar{\varrho}_i, \\ \tilde{\kappa}_i\hat{\zeta}_i^T\Omega\hat{\zeta}_i, & \varrho_i < \bar{\varrho}_i, \end{cases} \tag{50}$$

where $\tilde{\kappa}_i$ and $\bar{\varrho}_i$ are positive constants. The event-triggered strategy is

$$t_{k+1}^i = \min\{t > t_k^i | \tilde{\psi}_i(t) \leq 0\}. \tag{51}$$

where $\tilde{\psi}_i(t)$ is a virtual internal variable. The updating law of $\tilde{\psi}_i$ is

$$\dot{\tilde{\psi}}_i = \begin{cases} -\tilde{\beta}_i & \|\varepsilon_i\| = 0, \\ -[\varrho_i^2\tilde{\psi}_i^2 + \tilde{\psi}_i + \phi_i]\frac{\varepsilon_i^T\Omega\varepsilon_i}{\varepsilon_i^TP\varepsilon_i} - \tilde{\beta}_i, & \|\varepsilon_i\| \neq 0, \end{cases} \tag{52}$$

Corollary 1. *If the undirected topology graph of the multi-agent system (48) meets Assumption 1, the fully distributed adaptive controller (49) with the dynamic triggering protocol (51) can make the system (48) reach consensus asymptotically when $\bar{\varrho}_i \geq 1$. Furthermore, the Zeno phenomenon can be removed with a positive minimum length of triggering intervals for each agent, which is shown as*

$$\tilde{\tau}_i = \frac{2}{\tilde{\Xi}}(\arctan\frac{2\Xi_1\bar{\tilde{\psi}}_i + \Xi_2}{\tilde{\Xi}} - \arctan\frac{\Xi_2}{\tilde{\Xi}}) > 0, \tag{53}$$

where $\tilde{\Xi} = \sqrt{4\Xi_1\Xi_3 - \Xi_2^2}$, $\Xi_1 = \frac{\bar{\varrho}_i^2\lambda_{\max}(\Omega)}{\lambda_{\min}(P)}$, $\Xi_2 = \frac{\lambda_{\max}(\Omega)}{\lambda_{\min}(P)}$ and $\Xi_3 = \frac{\bar{\tilde{\psi}}_i\lambda_{\max}(\Omega)}{\lambda_{\min}(P)} + \tilde{\beta}_i$. $\bar{\tilde{\psi}}_i$ is the initial value of $\tilde{\psi}_i$.

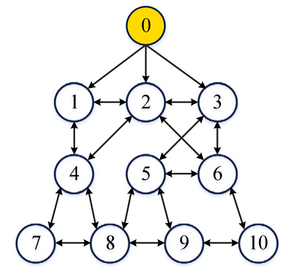

Fig. 1. Communication graph corresponding to the leader-following multi-agent system in Example 1.

Corollary 1 can be proofed with similar process of Theorems 1 and 2, which is omitted here.

Remark 6. Compared with the convergence conditions obtained in [32,39], the convergence conditions with lower conservatism are given in this work. Then, the positive minimum length of triggering intervals for each agent is obtained to eradicate Zeno behavior, making it easier to implement in engineering. Moreover, the number of triggering can be reduced by adjusting related parameters.

4. Simulation experiments

Example 1. Choose a multi-agent system with 10 followers as the first research object. Its topology is shown as Fig. 1, where the agent marked yellow is the leader and others are followers. Obviously, there exist edges that point from the leader to followers and Assumption 2 is satisfied. To satisfy Assumption 1, system matrices can be set as $D = [0, 1, 0; -6, 0, 4; 8/3, 0, -8/3]$ and $E = [0, 2, 0]^T$.

According to the algebraic Riccati Eq. (3) and setting $F = I$, one can obtain $P = [1.9259, -0.0128, -0.2162; -0.0128, 0.4936, 0.3788; -0.2162, 0.3788, 0.6481]$. As the initial value of $\dot{\psi}_i$, $-\beta_i$ can make the value of ψ_i have a downward trend after each triggering. So it can be randomly set as $-\beta_i = -10^{-6}$. Initial virtual internal variables are randomly set as $\bar{\psi}_i = 40$ for $i = 1, 2, \dots, 10$. Let $\chi_i(0) = 0.3$ and $\bar{\chi}_i = 1 + 0.1 * i$, $i = 1, 2, \dots, 10$. The convergence condition in Theorem 1 is satisfied. Initial followers' states are $z(0) = [-10, -8, -6, -4, -2, 1, 13, 25, 37, 49; -5, -7, -1, -4, -2, 0, 3, 5, 8, 9; -0.1, 0.2, -0.3, 0.4, 0.5, -0.6, 0.7, -0.8, 0.9, 1]$. Initial leader's state is set as $z_0(0) = [20, -20, 1]^T$.

State trajectories of different dimensions for all agents are shown in Fig. 2. One can see that the states of all followers gradually approach the state of the leader as time goes on and eventually become the same as the state of the leader. Thus, the tracking consensus is achieved under the proposed controller (7) and event-triggered mechanism (10). Fig. 3 shows all triggering instants t_k^i of each follower. Since the real-time state information only needs to be transmitted to neighboring agents at triggering instants, it can be seen from Fig. 3 that the event-triggered strategy effectively reduces the communication costs. The evolutions of adaptive parameters χ_i and ϕ_i are shown in Fig. 4 and Fig. 5, respectively. As time goes on, χ_i and ϕ_i turn to be constant values. This also shows that the system state is gradually becoming stable. As an example for studying the triggering interval, the actual minimum length of triggering intervals for 7th follower is $1.5110s$. According to the formula in Theorem 2, the minimum length of triggering intervals is $0.1185s$, which is smaller than the actual one obtained in the above example. The current result shows that the length of triggering intervals can be lower bounded by the one calculated according to Theorem 2.

Example 2. Take a leaderless multi-agent system with 6 agents as an example. Fig. 6 shows its topology. $D = [0, 1, 0; 0, 0, 1; 0, 0, 0]$ and $E = [0, 0, 1]^T$. Obviously, Assumptions 1 and 2 are satisfied.

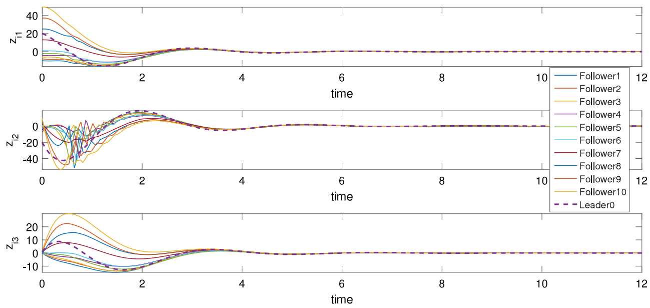

Fig. 2. State trajectories of agents in Example 1.

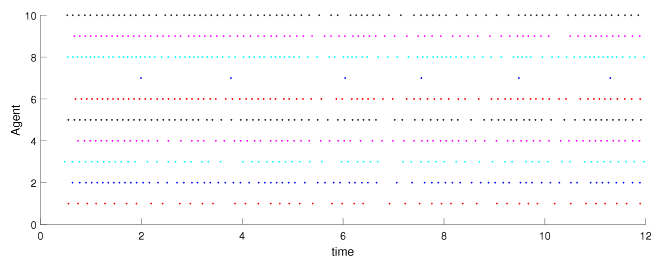

Fig. 3. Triggering instants of each follower in Example 1.

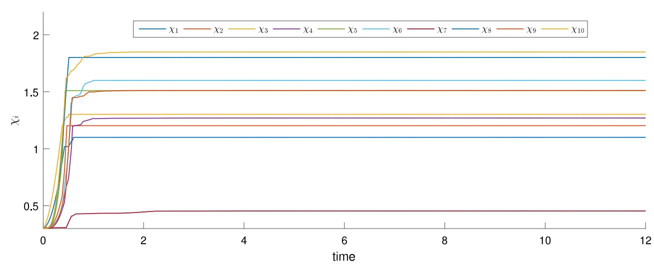

Fig. 4. Values of the adaptive parameters χ_i in Example 1.

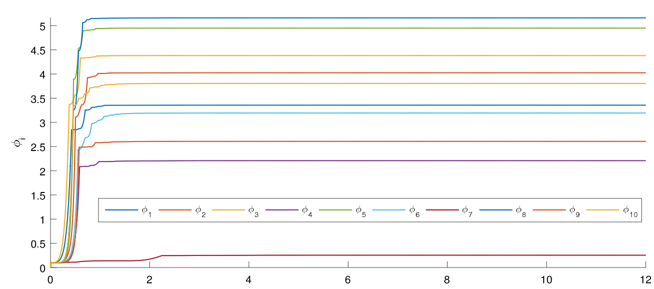

Fig. 5. Values of the adaptive parameters ϕ_i in Example 1.

According to the algebraic Riccati Eq. (3) and setting $F = I$, one obtains $P = [2.4142, 2.4142, 1.0000; 2.4142, 4.8284, 2.4142; 1.0000, 2.4142, 2.4142]$. As the initial value of $\dot{\tilde{\psi}}_i$, it is randomly set as $-\tilde{\beta}_i = -10^{-6}$. Initial virtual internal variables are randomly set as $\bar{\tilde{\psi}}_i = 70$ for $i = 1, 2, \ldots, 10$. Assume $\varrho_i(0) = 0.1$ and $\bar{\varrho}_i = 1+0.1 * i$, $i = 1, 2, \ldots, 10$. The convergence condition in Corollary 1 is satisfied. Initial agents' states are randomly set as $z(0) = [-5, 3, -2, 3.5, 1.7, -1.3; 4, 1.7, 1.1, 2.6, 1.2, 1; -2, 1.2, -0.8, -3, 1.5, 2.2]$.

Fig. 7 shows the state evolutions of different dimensions for all agents. Obviously, the state values of all agents gradually approach each other over time and eventually become the same value. Then, one can conclude that the leaderless consensus is achieved under the proposed controller (49) and event-triggered mechanism (51). Fig. 8 depicts all triggering instants t_k^i of each agent, which shows the contribution of the proposed method in reducing communication burdens. The evolutions of adaptive parameters ϱ_i and ϕ_i are the same as them in Example 1, which are not shown here. As an example for studying the length of triggering interval, the actual minimum length of triggering intervals for 5th agent is $0.2370s$. According to the formula in Corollary 1, the minimum length of triggering intervals is $0.0289s$, which is smaller than the actual one obtained in the above example. The current result shows that the length of triggering intervals can be lower bounded by the one calculated according to Corollary 1.

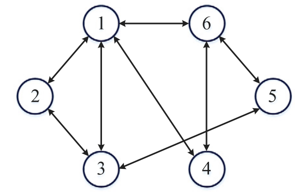

Fig. 6. Communication graph corresponding to the leaderless multi-agent system in Example 2.

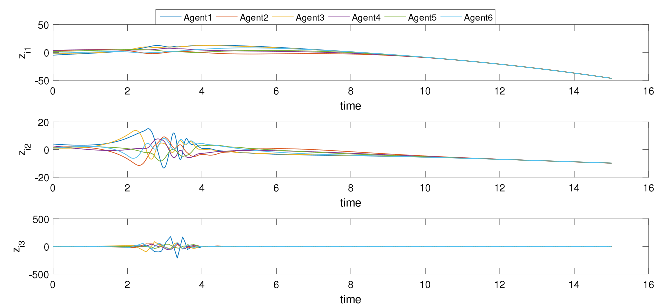

Fig. 7. State trajectories of agents in Example 2.

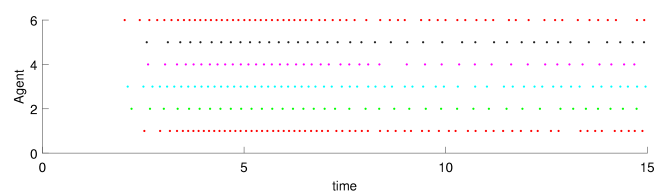

Fig. 8. Triggering instants of each agent in Example 2.

To make a comparison, experiments are also conducted with the fully distributed event-triggered strategy in [34] (Strategy 1) and the dynamic event-triggered strategy in [32] (Strategy 2). Table 1 shows triggering numbers corresponding to each agent under different event-triggered strategies. By comparing the triggering numbers corresponding to the same agent in the same time period under different event-triggered strategies, we can see that strategy 3 has a great advantage in reducing triggering numbers. This means that, compared with the other two event-triggered strategies, the strategy proposed in this paper has a great advantage in saving communication resources.

5. Conclusion

Fully distributed consensus control algorithms for general linear multi-agent systems by dynamic event-triggered approach have been studied in this note. First, to reduce communication burdens and avoid global features of multi-agent networks, a fully distributed adaptive event-triggered controller based on the open-loop state estimator of agents was addressed. Then, a novel triggering mechanism with a dynamic variable was designed, which can rule out Zeno behavior

Table 1
Comparison of triggering numbers under different event-triggered strategies in Example 2.

Agent i		1	2	3	4	5	6
0–5 s	Strategy 1	25	16	41	23	27	32
	Strategy 2	52	34	47	34	39	34
	Strategy 3	16	10	16	11	9	18
5–10 s	Strategy 1	47	14	38	29	30	35
	Strategy 2	67	19	53	18	16	37
	Strategy 3	30	16	25	19	17	28
10–15 s	Strategy 1	74	34	54	29	18	81
	Strategy 2	100	20	99	26	20	153
	Strategy 3	22	13	19	13	14	18

through ensuring a positive minimum length of triggering intervals. It has inferred that the convergence of multi-agent systems can be guaranteed asymptotically with lower conservative conditions. These advantages make it feasible to apply the proposed algorithm into engineering practices. Extending these strategies to general linear multi-agent systems with directed graphs, cyber attacks and nonlinear multi-agent systems are future works.

CRediT authorship contribution statement

Tongtong Chen: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. **Fuyong Wang:** Investigation, Software, Validation, Writing – original draft. **Meiling Feng:** Visualization, Investigation. **Chengyi Xia:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing. **Zengqiang Chen:** Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

No data was used for the research described in the article.

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